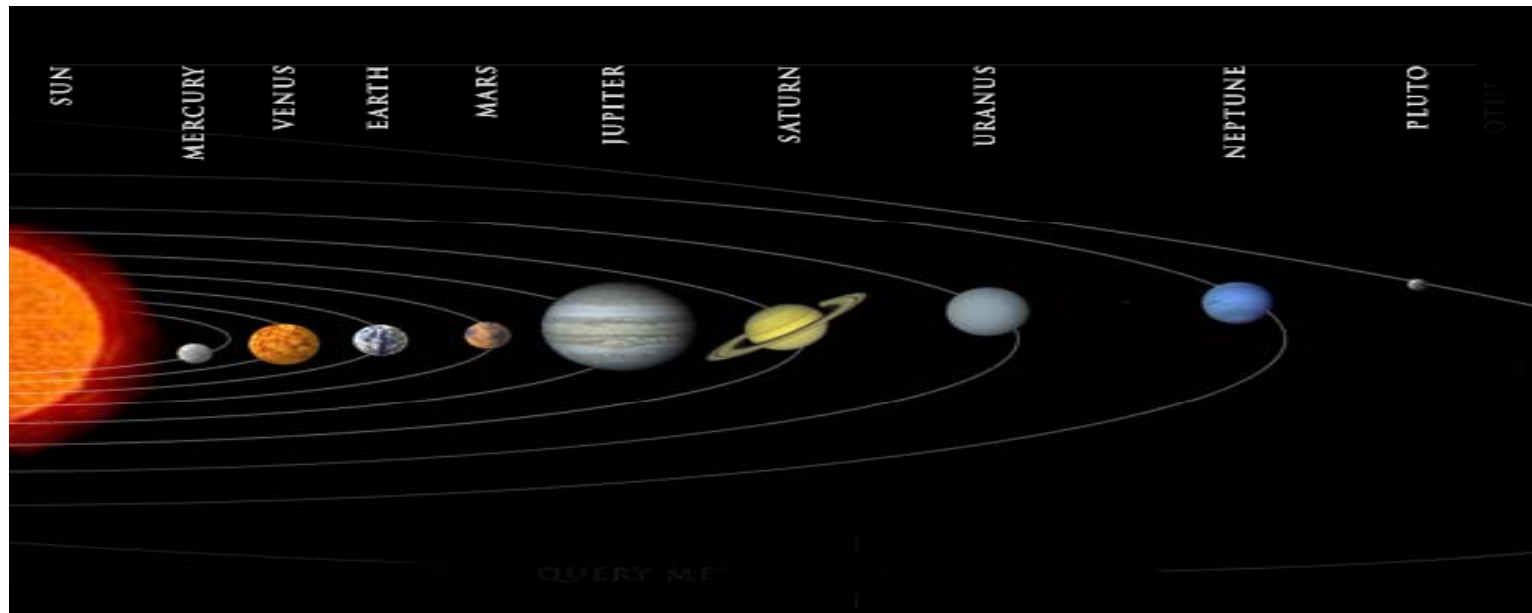




Selfstudy.in

My Class Notes

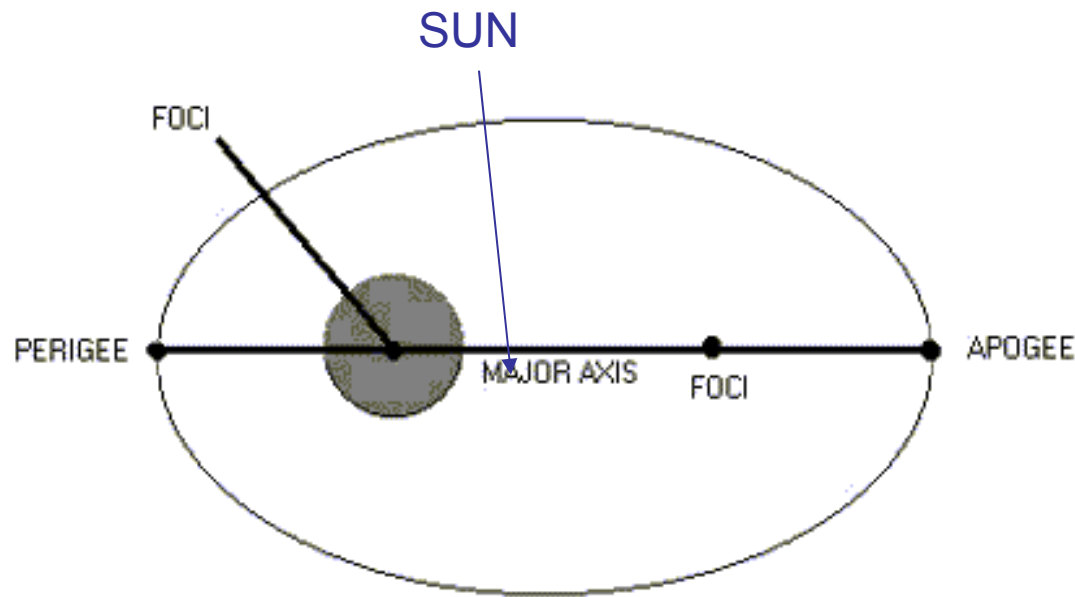
My Class Kepler's Laws



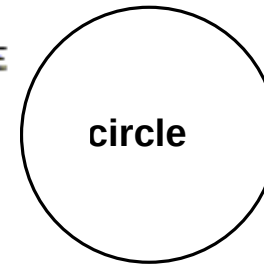
Kepler's Three Laws of Planetary Motion



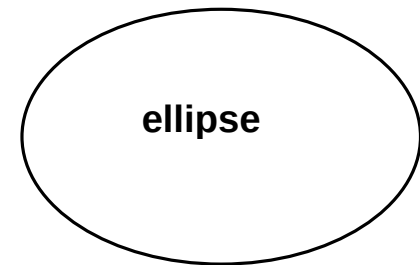
First Law: Planets move in **elliptical** orbits with the sun at one of the **foci**



Foci – 2 points that are equidistant from center on major axis



circle

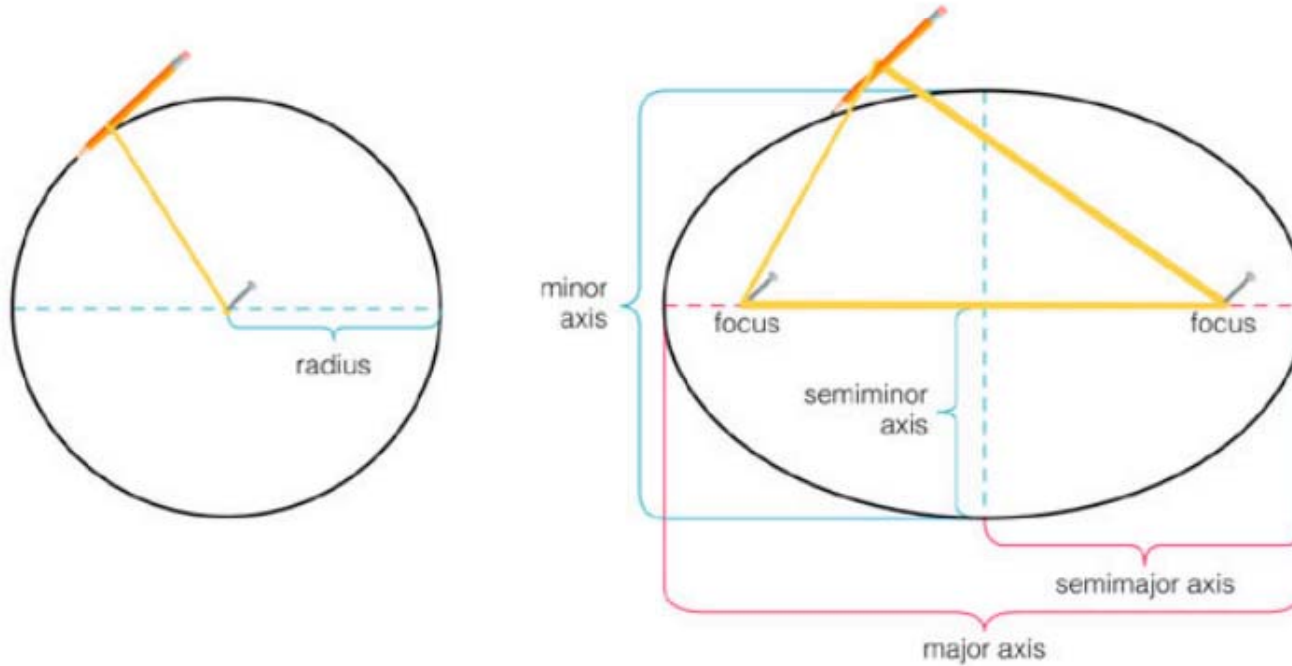


ellipse

An ellipse is a geometric shape with 2 foci instead of 1 central focus, as in a circle. The sun is at one focus with nothing at the other focus.

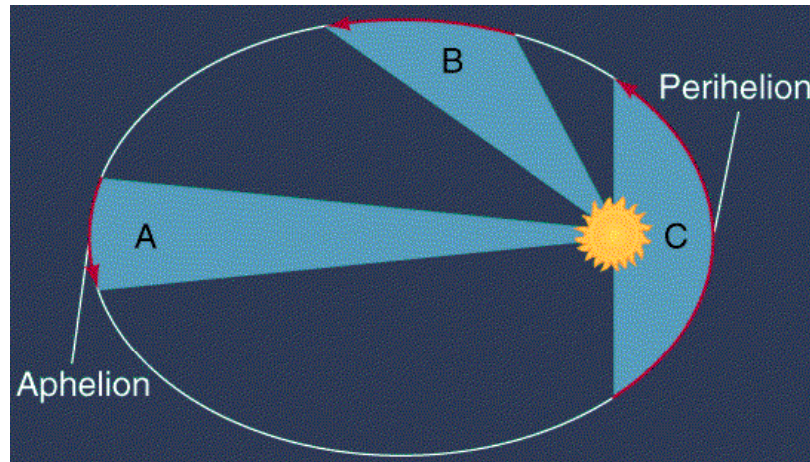


Geometry of ellipses

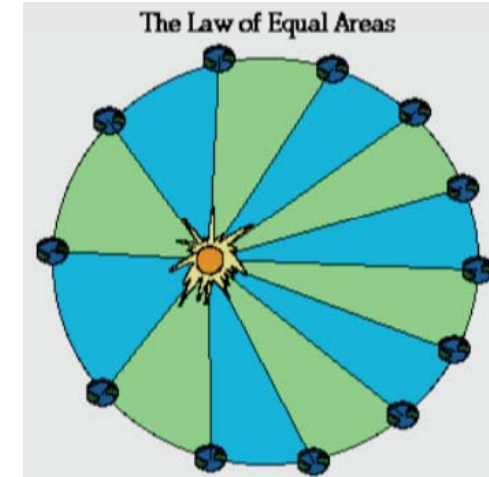




Second Law : Law of Equal Areas: Line from the sun to a planet sweeps with **equal** areas in **equal** time. A planet will move through **equal** area of space in an **equal** amount of time

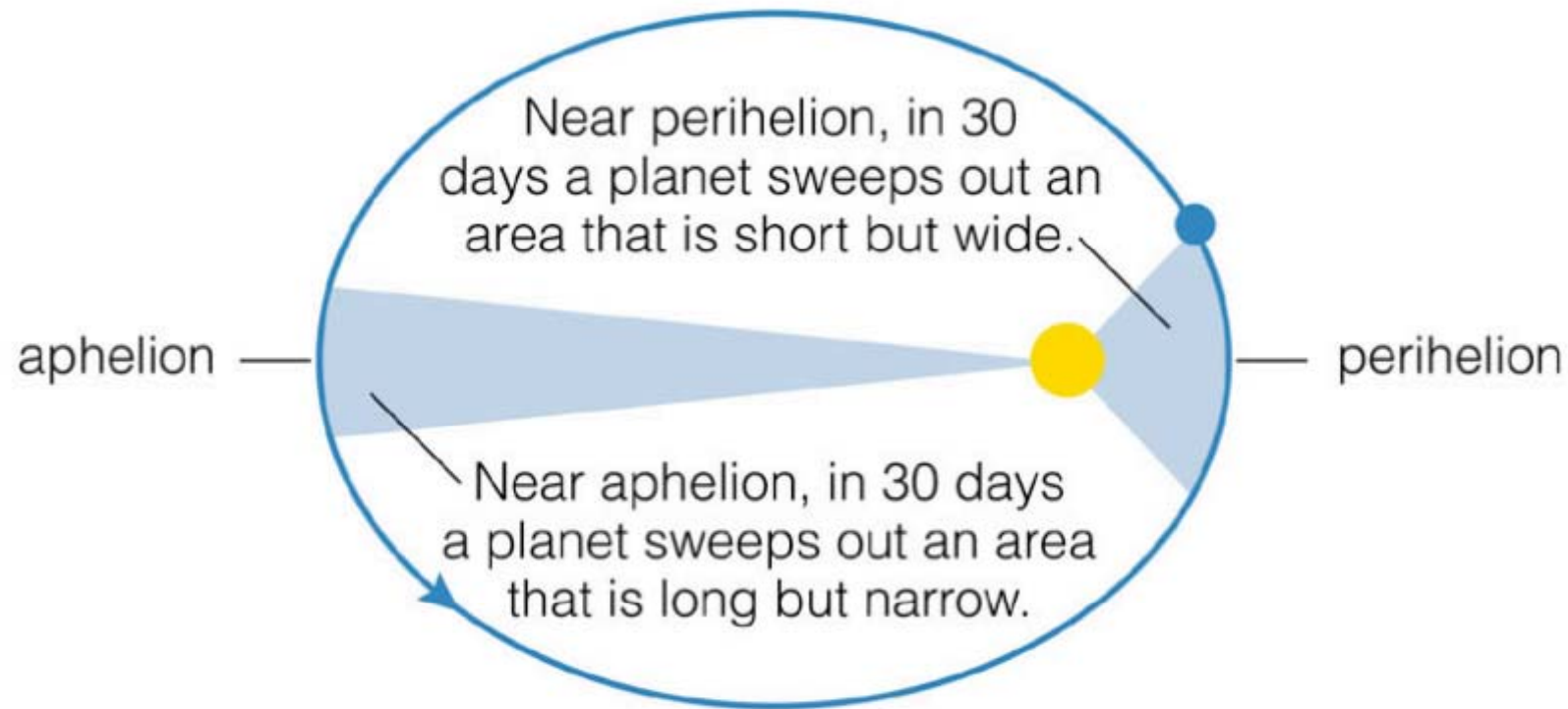


Perihelion – closest to sun
Ahelion – farthest from sun



When a planet is in perihelion its orbital velocity increases

When a planet is in aphelion its orbital velocity decreases

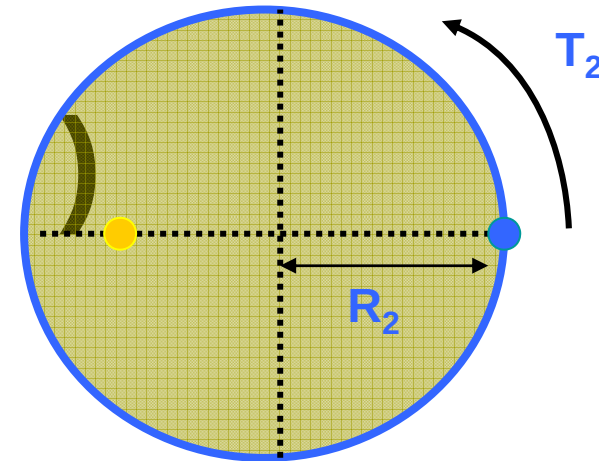
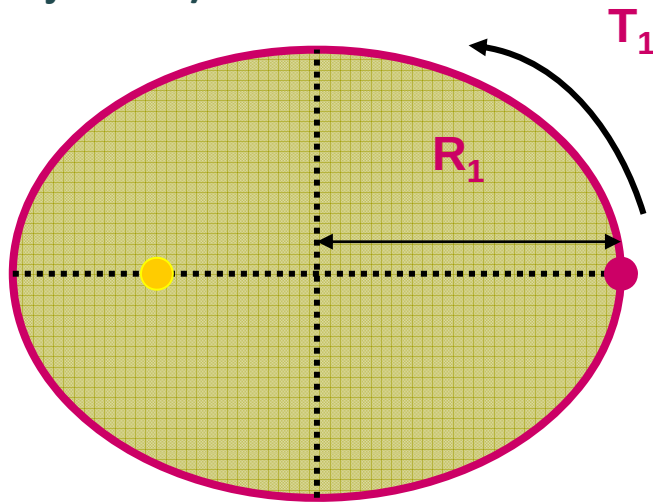


The areas swept out in 30-day periods are all equal.



3rd Law of Periods - The farther a planet is from the focus, the **longer** the period of revolution.

Kepler found a relationship between the time it took a planet to go completely around the sun (T , sidereal year), and the average distance from the sun (R , semi-major axis)



$$\frac{T_1^2}{T_2^2} = \frac{R_1^3}{R_2^3}$$



Kepler's third law: $P^2 \propto a^3$ Therefore $P^2 = ka^3$,

where k is some constant number.

We can find k if P is expressed in (Earth's orbit) years and a is expressed in terms of the distance between Sun & Earth.

This distance is called 1 Astronomical Unit, or 1 A.U.

Then substituting above values into $P^2 = ka^3$

we find: $(1)^2 = k(1)^3$. Solving for k , we find $k = 1$. When these units for P and a are used